SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

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Weightage:	20%	**Total Marks:**	25
Student Name:	Vigneshwaran R	**Reg. No.:**	1911260109
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Code of Conduct

I Vigneshwaran R / 1911260109 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Problem Statement 31: Smart Municipal Complaint Management System

1. Understanding the Problem Statement

Problem in my own words. Citizens face civic problems such as damaged roads, overflowing garbage bins, streetlight failures, drainage blockages and water supply issues. They report these through phone calls, letters or visits to different offices, and there is no central place where complaints are recorded and followed. Complaints get lost, are sent to the wrong department, or stay unresolved without any update to the citizen. A system is needed where citizens can report an issue with a description, photograph and location, and where the complaint is automatically categorized, routed to the right department and tracked until it is resolved.

Objectives of the proposed system

- Enable citizens to register complaints with description, photo and location.
- Automatically categorize complaints and assign them to the appropriate departments.
- Track resolution progress and maintain complete complaint histories.
- Notify citizens about every status update.
- Generate municipal performance analytics and reports.

Scope

In scope: a web and mobile application for citizens, department officers and administrators covering complaint registration, location capture, automatic categorization and routing, status tracking, notifications, escalation, history and analytics for one municipality. *Out of scope:* the physical repair work itself, payment of taxes and bills, and emergency services such as police and fire response.

2. Key Stakeholders and Challenges

Stakeholder	Roles and responsibilities	Key challenges
Citizens	Register complaints, add photos and location, track status, give feedback	No single channel, no updates, unsure whether anyone is working on the issue
Department Officers (roads, sanitation, lighting, drainage, water)	Receive assigned complaints, plan work, update status and close complaints	Wrong or late assignment, unclear location, heavy workload
Field Staff / Technicians	Visit the site, carry out repairs, upload completion proof	Poor location details, no work orders, manual reporting
Municipal Administrator / Commissioner	Monitors performance, reviews delays, allocates resources	No centralized data, weak accountability, delayed decisions
System Administrator	Manages users, departments, categories and security	Data security, system availability, misuse and duplicate complaints
Elected Representatives (indirect)	Review civic issues in their areas	Lack of reliable data on pending issues

Major challenges: lost or delayed complaints, misrouting between departments, duplicate complaints, inaccurate location details, no status updates for citizens, poor accountability, and uneven allocation of staff and resources.

3. Analysis of the Current Approach

Existing process. Citizens visit municipal offices, write applications, call helpline numbers or contact officials directly. Clerks note complaints in registers or spreadsheets and forward them manually to departments. Follow-up is done by phone, and reports are prepared by hand.

Strengths	Limitations
Familiar to citizens and staff Face-to-face contact is possible No technology cost Works for people without smartphones	Complaints are not tracked centrally Manual categorization and routing cause delays No photo or exact location No status notifications to citizens Duplicate and lost complaints No escalation of delayed cases No analytics for performance or planning

Gaps to be addressed: no centralized complaint registry; no automatic categorization and assignment; no location and photo evidence; no progress tracking or citizen notifications; no escalation of delays; and no performance analytics.

4. Proposed Solution Using Software Engineering Principles

Solution design. A layered web and mobile application. A citizen submits a complaint with text, a photo and GPS location picked on a map. A categorization engine classifies the complaint (road, garbage, streetlight, drainage, water supply) from its text and image using keyword rules and a trained text classifier, then routes it to the department responsible for that category and ward. Officers receive it in a dashboard, assign it to field staff and update status. Notifications go to the citizen by SMS, email or app. A React and Flutter front end, a Node.js or Spring Boot REST backend with JWT authentication, a PostgreSQL database with geospatial support, a map service and cloud file storage for photos form the stack.

Major functional modules

Module	Description
User and Role Management	Citizen registration, login, officer and admin roles
Complaint Registration	Description, photo upload, map-based location capture, duplicate detection
Automatic Categorization	Classify the complaint by type and priority
Department Assignment	Route to the right department and ward; reassign if needed
Resolution Tracking	Status stages: registered, assigned, in progress, resolved, closed; proof of work
Notification Module	SMS, email and in-app updates to the citizen at every stage
Complaint History and Escalation	Full history per complaint; automatic escalation of delayed cases
Analytics and Reports	Resolution time, pending counts, department performance, hotspot maps

Functional requirements

- Register and log in as citizen, officer or administrator.
- Submit complaints with description, photograph and location.
- Automatically categorize and assign complaints to departments.
- Allow officers to update status and upload completion proof.
- Notify citizens of each status change.
- Let citizens track complaints and view history; collect feedback.
- Generate dashboards and performance reports.

Non-functional requirements

- **Performance:** complaint submission within 3 seconds; dashboards load in under 5 seconds.
- **Availability:** at least 99% uptime.
- **Security:** encrypted communication, role-based access, protection of citizen data.
- **Usability:** simple, multilingual, mobile-friendly interface.
- **Scalability:** supports thousands of complaints per day across many wards.

Software engineering principles applied

Principle	How it is incorporated
Modularity	Independent modules (registration, categorization, assignment, tracking, notification, analytics) with defined APIs
Scalability	Stateless services, message queue for notifications, database indexing, cloud deployment
Maintainability	Clean code, documentation, Git version control, automated tests, configurable categories and rules
Reliability	Input validation, retry for failed notifications, backups, escalation so no complaint is left unattended
Usability	Simple report form, map pin selection, status timeline, local language support
Security	JWT authentication, role-based access control, HTTPS, image validation, audit logs, privacy of citizen data

5. Justification of the Chosen Methodology: Agile (Scrum)

Selected framework: Agile using Scrum, with two-week sprints, a product backlog, sprint reviews and retrospectives.

Why it is suitable

- Needs of citizens and departments vary and will change after pilot use; Scrum accepts changing requirements.
- The system can be delivered incrementally: Sprint 1 login and complaint registration; Sprint 2 categorization and department assignment; Sprint 3 tracking and notifications; Sprint 4 escalation, analytics and reports.
- A working version can be piloted in one ward early, giving real feedback from citizens and officers.
- Waterfall is too rigid for a citizen-facing system that evolves, and Spiral adds more risk-analysis overhead than needed.

Support for development and maintenance

- Sprint reviews with municipal staff keep the product aligned with real workflows.
- Continuous integration and testing protect routing and notification logic.
- The categorization model and rules can be improved in later sprints.
- After release, new departments, categories and languages are added as backlog items.

6. Expected Outcomes and Limitations

Expected benefits

- Faster resolution of civic issues.
- Improved municipal accountability and transparency.
- Better citizen communication through status updates.
- Centralized complaint tracking and history.
- Improved resource allocation and increased citizen satisfaction.

How the solution addresses the challenges

Challenge	Solution
Lost or scattered complaints	One centralized registry with unique complaint IDs
Wrong or late routing	Automatic categorization and department assignment
Unclear location	Map-based GPS location and photo evidence
No updates for citizens	Automatic notifications at every status change
Poor accountability and planning	Escalation rules and performance analytics

Risks and limitations

- Automatic categorization may be wrong for unclear or mixed complaints; officers need to correct them.
- Citizens without smartphones or internet need assisted channels such as helplines.
- Fake or duplicate complaints and misuse must be controlled.
- Depends on departments updating status honestly; staff training and acceptance are needed.

Future enhancements

- Image-based automatic issue detection and severity estimation.
- Voice and chatbot complaint registration in local languages.
- Predictive hotspot analysis for preventive maintenance.
- Integration with smart city IoT sensors and field staff vehicle tracking.